

Final Project Report

Date	20 th July 2026
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Project Name	Food Ordering Behaviour Analysis

1. Introduction

1.1. Project Overview

Food Ordering Behaviour Analysis is a data analytics project that turns raw food-delivery transaction records into an interactive Tableau dashboard and story. The dataset covers approximately 50,000 orders across six cities (Bangalore, Chandigarh, Delhi, Hyderabad, Mumbai, Pune), six cuisine types (Biryani, Chinese, Desserts, Fast Food, North Indian, South Indian), four meal types (Breakfast, Lunch, Dinner, Snacks), and company/dining segments (Alone, Family, Friends, Partner). The dashboard is embedded into a lightweight Python Flask web application and published for public access via Tableau Public.

1.2. Objectives

- Consolidate fragmented food ordering data (city, cuisine, meal type, ratings, delivery time, revenue) into a single interactive analytical view.
- Identify which cities, cuisines, and meal times drive the highest order volume and revenue.
- Surface delivery-time performance and customer rating patterns to support operational improvements.
- Publish a publicly accessible, interactive Tableau Story that communicates consumer ordering trends to non-technical stakeholders.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

- **PS-1 (Business Analyst):** trying to explore real-time data visualisations of cuisine demand and consumer habits across food ordering platforms, but order volume, ratings, delivery time, and revenue are tracked as separate, disconnected views, because there is no single interconnected dashboard linking overall corporate KPIs together — which makes it hard to see a structured, unified picture of food ordering behaviour and consumer trends.
- **PS-2 (City Operations Manager):** trying to identify which cities show strong order volume alongside consistent delivery performance, but order volume by city and delivery time distribution are shown as separate charts, because there is no combined view linking cities, delivery time, and order type together — which makes it hard to tell whether growth in a city is matched by reliable delivery service.

2.2. Project Proposal (Proposed Solution)

- **Objective:** Develop, integrate, and deploy an interactive Tableau-based dashboard translating food-ordering trends into accessible, shareable visual insights.
- **Approach:** Raw order data is cleaned and structured, visualized in Tableau Public as a multi-layered Story (KPI cards, bubble maps, bar charts, histograms), and embedded into an HTML frontend served by a Python Flask application.
- **Key Features:** A unified narrative merging KPI panels; cloud-native delivery via Tableau Public / Flask; interactive filtering by city, cuisine, meal type, and company/dining segment.

2.3. Initial Project Planning

Product backlog organized across two sprints:

- Sprint-1: Data collection & preprocessing of the food ordering dataset; dashboard design for KPI/cuisine breakdown and rating/city analysis.
- Sprint-2: Dashboard design for delivery time and company-type analysis; assembly of the interconnected Story linking overall KPIs.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data source: food_ordering_behavior_dataset.csv, sourced from the project's GitHub repository (Food-Ordering-Analysis-Flask), in CSV format, containing order-level records with city, cuisine, meal type, company/dining segment, rating, delivery time, and order value.

3.2. Data Quality Report

- **Issue identified:** Inconsistent text formatting — double spacing, irregular casing, and trailing whitespace in categorical fields such as City, Cuisine, and Meal Type (Moderate severity).
- **Resolution:** Applied string trimming and case normalization so identical values group correctly in Tableau.

3.3. Data Exploration and Preprocessing

- Data Cleaning: removed incomplete records and duplicate Order IDs; normalized categorical text fields.
- Data Transformation: built calculated fields for average rating and average delivery time by meal type/city; aggregated Count of Order Id and revenue by meal type.
- Data Type Conversion: Order Id as string identifier; Order Value and Rating as floats; Time Taken to Order as numeric.
- Data Modeling: linked the orders table to City, Cuisine, Meal Type, and Company Type dimensions via Order Id.
- Save Processed Data: exported the standardized dataset as food_ordering_behavior_dataset.csv for connection to Tableau Public.

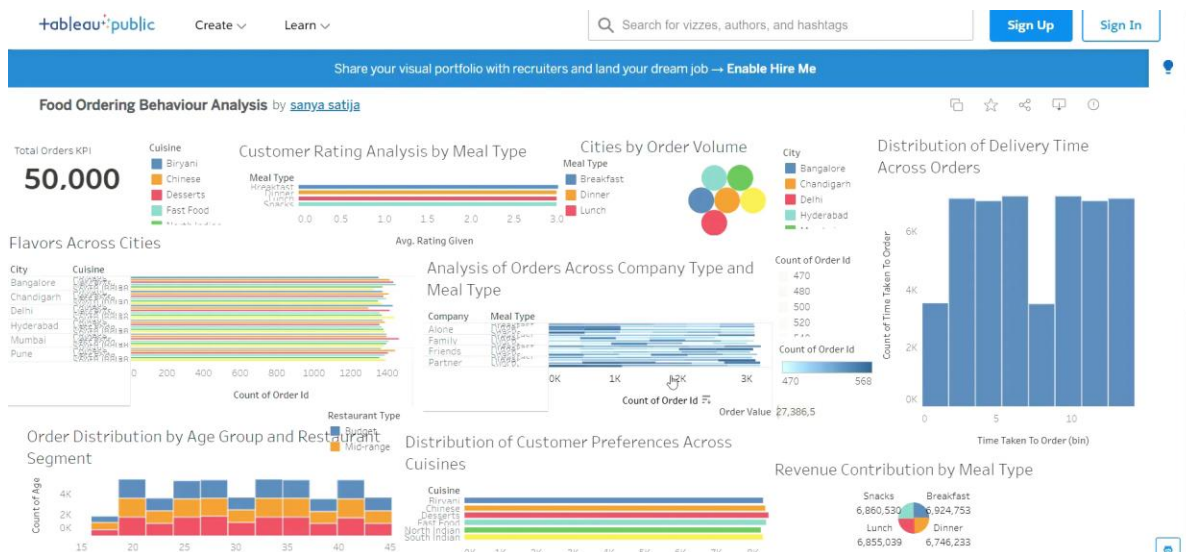
4. Data Visualization

4.1. Framing Business Questions

- Which cities generate the highest order volume?
- How do customer ratings vary across meal types?
- What does the distribution of delivery time look like across all orders?
- How does cuisine preference differ across cities?
- How are orders distributed across company/dining type and meal type?
- How is order volume distributed by customer age group and restaurant segment?
- How is total revenue contributed across the four meal types?
- What is the distribution of customer preferences across cuisines overall?

4.2. Developing Visualizations

Visualizations built: a packed bubble chart (Cities by Order Volume), horizontal bar chart (Customer Rating Analysis by Meal Type), histogram (Distribution of Delivery Time Across Orders), stacked bar chart (Flavors Across Cities), grouped bar chart (Orders by Company Type and Meal Type), column chart (Order Distribution by Age Group), and a revenue-contribution breakdown by meal type.

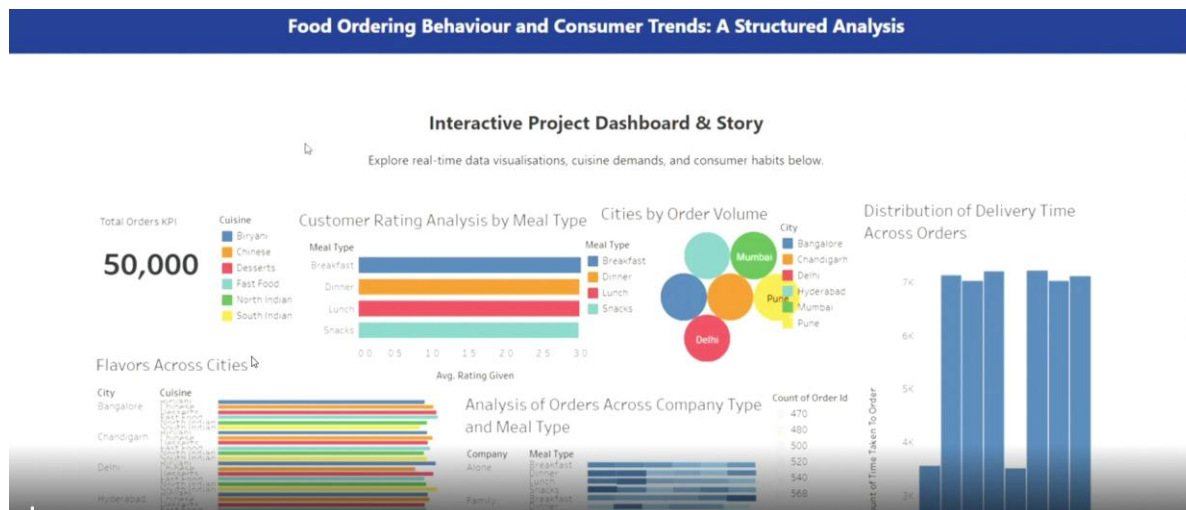


5. Dashboard

5.1. Dashboard Design File

The dashboard applies a clear and intuitive layout, consistent color theming by category, interactive filters, and responsive design. Key outcomes:

- Regional order concentration across the six tracked cities highlights where operational and delivery resourcing should be prioritized.
- Group dining (Family & Friends) drives a large share of orders at specific meal times.
- Cuisine demand is unevenly distributed across cities, supporting city-tailored menu strategy.
- Lunch and Snacks contribute higher revenue shares than Breakfast.



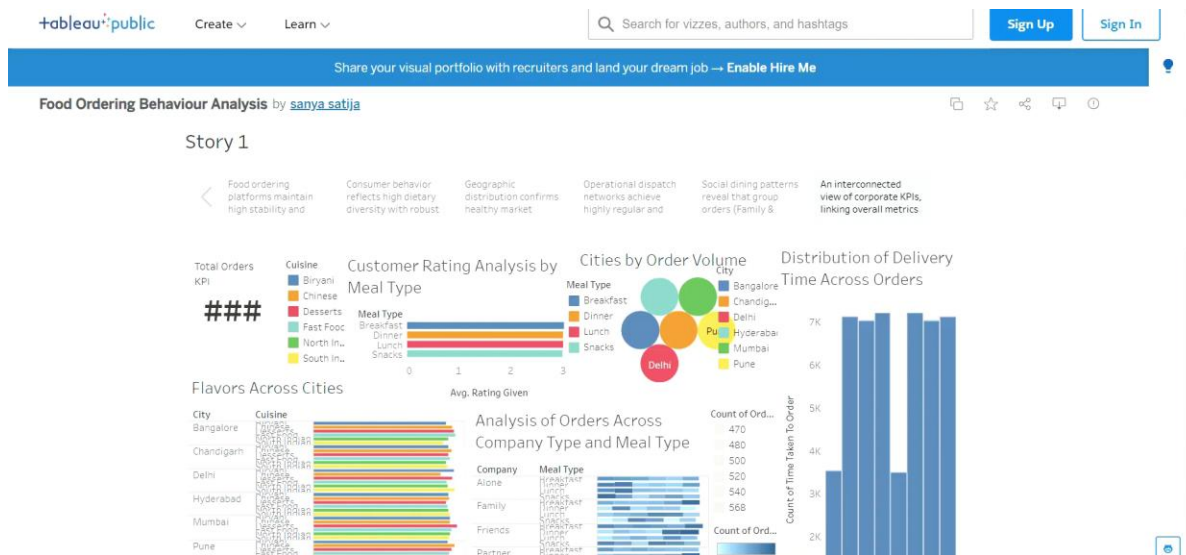
6. Report

6.1. Story Design File

The Tableau Story, titled “Food Ordering Behaviour and Consumer Trends: A Structured Analysis,” guides viewers through a sequential narrative:

- Food ordering platforms maintain high stability.
- Consumer behavior reflects high dietary diversity.

- Geographic distribution confirms a healthy market across all six cities.
- Operational dispatch networks achieve highly regular delivery timing.
- Social dining patterns reveal those group orders (Family & Friends) make up a substantial share of totals.
- An interconnected view of corporate KPIs links overall metrics into one unified story.



7. Performance Testing

Metric	Observation (from the published dashboard)
7.1 Utilization of Data Filters	4 interactive filters/legends used — City, Cuisine, Meal Type, and Company Type — enabling cross-filtering across all sheets in the dashboard.
7.2 No. of Calculation Fields	Approximately 4–5 calculated fields — including average rating, average delivery time, Count of Order Id, and revenue contribution by meal type.
7.3 No. of Visualizations	10 visualizations across the dashboard and story: Total Orders KPI, Cuisine legend, Customer Rating Analysis, Cities by Order Volume, Delivery Time Distribution, Flavors Across Cities, Orders by Company Type & Meal Type, Order Distribution by Age Group, Customer Preferences by Cuisine, and Revenue Contribution by Meal Type.

8. Conclusion/Observation

The Food Ordering Behaviour Analysis dashboard successfully consolidates city-wise order volume, cuisine demand, delivery performance, customer ratings, and revenue into a single interactive Tableau Story. The analysis confirms healthy, diversified demand across cities and cuisines, consistent delivery performance, and a strong contribution from group dining and lunch/snack orders to overall revenue.

9. Future Scope

- Incorporate real-time order feeds to move from static/periodic data refresh to live monitoring.
- Add predictive analytics (e.g., demand forecasting by city/cuisine/meal-time) to support inventory and staffing decisions.
- Extend the dashboard with customer-level segmentation (repeat vs. one-time customers) for retention analysis.
- Integrate delivery-partner performance metrics to correlate delivery time with driver-level operational data.